Solutions to Chapter 12 Exercises

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

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Notation. For a hyperbolic periodic orbit of period k: S = stable, U^+ = regular unstable ($(f^k)' > 1$), U^- = flip unstable ($(f^k)' < -1$). **Facts used.** (i) $G(x) = 4x(1-x)$ and $f_2(x) = 2 - x^2$ are conjugate (by $C(x) = 4x - 2$, Exercise 3.3) to the tent map T_2, so every periodic orbit of period k has multiplier $\pm 2^k$, with sign $(-1)^r$, where r is the number of points of the orbit in the right half (the decreasing branch, symbol R). (ii) The logistic family is conjugate to the quadratic family: $g_a(x) = ax(1-x)$ is conjugate, via $y = a(x - \frac{1}{2})$, to $f_c(y) = c - y^2$ with $c = a(a-2)/4$, increasing in a for $a > 1$.

Exercise 12.1

Take $a_0 = -1$, $a_c = 2$ and the region $V = [-2, 2]$.

1. *At $a_0 = -1$:* $f_{-1}(x) - x = -(x^2 + x + 1) < 0$ for all x, so every orbit is strictly decreasing; there are no fixed or periodic points at all (allowed by Hypothesis 1).

2. *At $a_c = 2$:* f_2 is conjugate to the tent map T_2 (by 3.3 and the conjugacy of G with T_2), so every periodic orbit of period k has multiplier $\pm 2^k$ and is a hyperbolic repeller (in U^+ or U^-); there are no stable periodic orbits, and periodic orbits exist (e.g. the fixed points 1 and -2).

3. *Periodic orbits stay in V:* for $a \in [-1, 2]$ and $|x| > 2$, $f_a(x) = a - x^2 < -2$ and $|f_a(x)| = x^2 - a \geq x^2 - 2 > |x|$ (because $x^2 - |x| - 2 = (|x| - 2)(|x| + 1) > 0$). So the orbit stays outside $[-2, 2]$ and moves strictly away: no periodic point lies outside V.

4. *Generic bifurcation hypotheses:* the period-k points are the zeros of the polynomial $f_a^k(x) - x$, and bifurcation orbits additionally satisfy $(f_a^k)'(x) = \pm 1$. Two polynomial conditions in (a, x) have only finitely many common solutions in the compact region $[-1, 2] \times V$ (they have no common curve of solutions), so for each k there are only finitely many bifurcation orbits. The nondegeneracy conditions that make every such bifurcation a generic saddle-node or period-doubling (as in Figures 12.7 and 12.10) hold for the quadratic family; checking them is the standard (more technical) part, which I take as known for this family.

Exercise 12.2

No period-two saddle node. For $f_a(x) = a - x^2$, $f_a^2(x) - x = (x^2 + x - a)(x^2 - x + 1 - a)$, so period-two points solve $x^2 - x + 1 - a = 0$, with roots $x_{1,2} = \frac{1}{2}\big(1 \pm \sqrt{4a-3}\big)$, distinct iff $a > \frac{3}{4}$. The multiplier is $f'(x_1)f'(x_2) = 4x_1x_2 = 4(1-a)$. A saddle node needs multiplier $+1$, i.e. $a = \frac{3}{4}$; but at $a = \frac{3}{4}$ the two roots coincide: the "period-two orbit" has collapsed onto the fixed point $\frac{1}{2}$ (this is the period-doubling point where $f' = -1$), so there is no genuine period-two orbit with multiplier 1. For $a > \frac{3}{4}$ the multiplier $4(1-a) < 1$. Hence no period-two saddle node. For the logistic family, the period-two multiplier is $-a^2 + 2a + 4$, which equals 1 only for $a = 3$ or $a = -1$, where again the two points merge into a fixed point ($a = 3$ is the period-doubling of the fixed point, and at $a = -1$ the roots $\frac{(a+1) \pm \sqrt{(a+1)(a-3)}}{2a}$ both equal 0). (Equivalently, use the conjugacy with f_c.)

A period-four saddle node must exist. At $a_c = 2$ (resp. $a = 4$ for the logistic family) there are $(2^4 - 2^2)/4 = 3$ period-four orbits, with itineraries $LLLR$, $LLRR$, $LRRR$. Their multipliers are $(-1)^r 2^4$ with $r = 1, 2, 3$ right-points respectively, so $LLRR$ is a regular repeller (U^+) and the other two are flip repellers. Consider the U^+ period-four orbit at a_c and follow its branch of period-four orbits to smaller a. By the structure of the proof of the Cascade Theorem, the branch of U^+ orbits cannot reach a_0 (no periodic orbits there), cannot leave V, and cannot end at a period-doubling orbit (a period-k branch of a snake ending in a period-doubling must be an S branch, Lemma 12.10). So it must end at a bifurcation orbit of period four of the other generic type: a **period-four saddle node**. (Consistently, the only period-two orbit at a_c is LR, a flip repeller, so no U^+ orbit of period two exists to lead to a period-two saddle node.) For the logistic family apply the same argument via the conjugacy with the quadratic family.

Exercise 12.3

Odd k for g_4. Period-k orbits of g_4 correspond one-to-one to cyclic equivalence classes of primitive binary words of length k in L, R, and the multiplier of the orbit is $(-1)^r 2^k$, where r is the number of R's. So the orbit is a regular repeller iff r is even and a flip repeller iff r is odd. The *complement* map (interchange $L \leftrightarrow R$) is an involution on the set of primitive cyclic words of length k and sends r to $k - r$. For k odd, r and $k - r$ have opposite parity, so the complement pairs each regular repeller with a flip repeller (and has no fixed points, since $r \neq k - r$). Hence exactly half of the period-k orbits are regular repellers and half are flip repellers.

Horseshoe. Periodic orbits of the hyperbolic horseshoe correspond to the same words, with one of the two strips flipping the unstable direction (as in 10.9, the right strip maps by $(1 - x/3, 3 - 3y)$ with unstable eigenvalue -3). The sign of the unstable eigenvalue of an orbit is $(-1)^r$, where r is the number of visits to the flipping strip; the same complement argument gives half regular saddles and half flip saddles for odd k.

Even k. The result does not hold in general. For even k, r and $k - r$ have the same parity, so the complement no longer pairs the two types. For example $k = 2$ has only the orbit LR ($r = 1$), a flip repeller: 0 regular and 1 flip. For $k = 4$ there are three orbits: $LLRR$ (regular) and $LLLR$, $LRRR$ (flip), so 1 regular and 2 flip. (The proportions do tend to $\frac{1}{2}$ as $k \to \infty$, by 12.4.)

Exercise 12.4

Let g be g_4 or the horseshoe; g^n has exactly 2^n fixed points, one for each binary word of length n, and the type (regular or flip) of the periodic point with word w is determined by the parity of $r(w)$.

(a) Of the 2^n words, exactly 2^{n-1} have r even and 2^{n-1} have r odd. Remove the N fixed points of g^n that are not of period n: each type loses at most N points, so there are at least $2^{n-1} - N$ period-n points of each type, among $2^n - N$ period-n points in all. Since each orbit consists of n points of one type, the proportion of orbits of one type is at least

$$\frac{2^{n-1} - N}{2^n - N} \;\geq\; \frac{2^{n-1} - N}{2^n} = \frac{1}{2} - \frac{N}{2^n}.$$

(b) The fixed points of g^n that are not of period n have period m dividing n with $m \leq n/2$, and there are at most 2^m fixed points of g^m. So

$$N \leq 2^{n/2} + 2^{n/3} + 2^{n/4} + \cdots + 2 \leq n\, 2^{n/2},$$

since there are fewer than n terms, each at most $2^{n/2}$.

(c) Let $h(x) = x2^{-x/2}$. Then $h'(x) = 2^{-x/2}\big(1 - \frac{x \ln 2}{2}\big) < 0$ for $x > \frac{2}{\ln 2} \approx 2.885$, so h is decreasing there. Hence for $n \geq 24$, $N/2^n \leq n2^{-n/2} = h(n) \leq h(24) = 24/2^{12} \approx 0.0059 < 0.01$.

Therefore for $n \geq 24$ the proportion of each type is at least $\frac{1}{2} - 0.01 = 0.49$: at least 49% of the period-n orbits are regular unstable and at least 49% are flip unstable.

Exercise 12.5

Let p be a flip-unstable (U^-) periodic orbit of period k at $a = a_c$.

(a) The set G and paths through bifurcation orbits. Let G consist of the U^- orbits of period k and the U^+ and S orbits of period j for all $j \geq 2k$.

- At a *saddle node* of period $j \geq 2k$, the two emanating branches are an S branch and a U^+ branch of period j: both lie in G, so G contains a path through the bifurcation orbit.
- At a *period-doubling* orbit of period $j \geq k$, three branches emanate: one of period $2j$ (type S or U^+, so in G because $2j \geq 2k$), and two of period j, one S and one U^-. If $j = k$, the branches in G are the U^- branch of period k and the period-$2k$ branch; if $j \geq 2k$, they are the S branch of period j and the period-$2j$ branch (the U^- branch of period $j > k$ and the S branch of period k are not in G). In each case G contains a path through the bifurcation orbit.

(b) New snakes. Orient the branches of G as $S \rightarrow$, $U^+ \leftarrow$, $U^- \leftarrow$ (arrow in the direction of travel along the snake: right means increasing a). Checking the generic pictures (Figures 12.7 and 12.10), the orientation is consistent at every bifurcation orbit of G: at a saddle node, an $S \rightarrow$ branch and a $U^+ \leftarrow$ branch lie on the same side, so the path arrives along U^+ and leaves along S (a turn); at

a period-doubling of period k the path goes $U^- \leftarrow$ then either $S(2k) \rightarrow$ (supercritical, a turn) or $U^+(2k) \leftarrow$ (subcritical, straight through), and at a period-doubling of period $j \geq 2k$, $S \rightarrow$ then $S(2j) \rightarrow$ (straight) or $U^+(2j) \leftarrow$ (turn). Now the lemmas:

- **12.10.** A period-j branch of a snake ending in a period-doubling orbit of period j is an S branch if $j \geq 2k$ and a U^- branch if $j = k$. This is read off the four generic period-doubling pictures (a regular unstable orbit never period-doubles, since for area-contracting maps and for scalar maps the derivative must pass through -1).

- **12.11 (no snake leaves $[a_0, a_c] \times V$).** Leaving through ∂V is excluded by Hypothesis 3. At $a = a_0$ only S orbits exist, oriented $\rightarrow$, i.e. pointing into the region. At $a = a_c$ there are only U^+ and U^- orbits, oriented $\leftarrow$, also pointing into the region. So a snake cannot leave.

- **12.12.** The snake through p has an endpoint at $a = a_c$ (the orbit p itself, which enters the region), so it is not a closed loop. It cannot leave the region (12.11), and by (a) and the generic hypotheses it cannot end at a bifurcation orbit or at an interior orbit (hyperbolic orbits continue, and each bifurcation orbit has a continuation in G). So it has infinitely many branches and bifurcation orbits.

- **12.13.** By Hypothesis 4 there are only finitely many bifurcation orbits of each period $\leq N$ in $[a_0, a_c] \times V$, so the snake has only finitely many of period $\leq N$.

- **12.14.** By 12.12 and 12.13 the snake contains bifurcation orbits of arbitrarily large period. Along a snake, the period changes only at period-doubling orbits ($j \rightarrow 2j$), never at saddle nodes. Starting from p (period k), the U^- branch of period k can only end at a period-doubling orbit of period k (saddle nodes join S and U^+ branches), so the snake passes through period-doubling orbits of periods $k, 2k, 4k, \ldots$, and by Lemma 12.10 it contains the stable branches (S) of periods $2k, 4k, 8k, \ldots$ attached to them. Finally the period-doubling orbit of period k also has an attached S branch of period k, which is therefore in the connected set of orbits (it is attached at that bifurcation orbit). So the connected set of orbits containing p contains stable branches of periods $k, 2k, 4k, \ldots$, all in the bounded region $[a_0, a_c] \times V$: a cascade from period k.